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# 1 Linear Algebra Review

In stats:

- Matrices: the dataset
- Vector: one variable of the dataset (we will always assume vectors to be vertical)
- Scalar: single value of a variable

Matrix Multiplication:

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}, B = \begin{pmatrix} 5 & 10 \\ 15 & 20 \\ 25 & 30 \end{pmatrix}, AB = \begin{pmatrix} (1*5+2*15+3*25) & (1*10+2*20+3*30) \\ (4*5+5*15+6*25) & (4*10+5*20+6*30) \end{pmatrix} = \begin{pmatrix} 110 & 140 \\ 245 & 320 \end{pmatrix}$$

$AB \neq BA$

Identity Matrix is just a diagonla matrix of all 1's

Basic Operations:

- Only square matrices have inverses
- Find $A^{-1}$ is long and slow
- $(A^{-1})^{-1} = A$
- $(sA)^{-1} = s^{-1}A^{-1}$ ($s$ is a scalar)
- $(AB)^{-1} = B^{-1}A^{-1}$
- If $D$ is diagonal, $D^{-1}$ is also diagonal with elements $\frac{1}{d_{ii}}$

The Determinant ($|A| = s$) corresponds to Variance of a dataset

Gradients:

$$f(x_1, x_2, x_3) = 3x_1^2 x_2 + x_2 x_3^3 - e^{x_1 x_2}, \nabla f(x) = \begin{pmatrix} 6x_1 x_2 - x_3 e^{x_1 x_3} \\ 3x_1^2 + x_3^3 \\ 3x_2 x_3^2 - x_1 e^{x_1 x_3} \end{pmatrix}$$